

```

Last login: Wed Aug 23 11:57:25 on console
ashwini@Ashwinis-MacBook-Air ~ % sudo pip3 install rdm6300
Password:
WARNING: The directory '/Users/ashwini/Library/Caches/pip/' or its parent directory is not owned or is not writable by the current user. The cache has been disabled. Check the permissions and owner of the directory. If executing pip with sudo, you should use sudo's -H flag.
Collecting rdm6300
  Downloading rdm6300-0.1.1-py3-none-any.whl (7.2 kB)
Collecting pyserial (from rdm6300)
  Downloading pyserial-3.5-py2.py3-none-any.whl (98 kB)
    98.6/98.6 kB 1.5 MB/s eta 0:00:00
Installing collected packages: pyserial, rdm6300
Successfully installed pyserial-3.5 rdm6300-0.1.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/en/latest/faq/#using-pip-as-root
ashwini@Ashwinis-MacBook-Air ~ %

```

Fig. 5: Installing library

```

RFIDread.py
1 import rdm6300
2 import csv
3 import datetime
4
5 reader = rdm6300.Reader('/dev/cu.usbserial-1110')
6 csv_filename = 'rfid_data.csv'
7 csv_header = ['RFID', 'Name', 'Date', 'Time']
8
9 # Write the CSV header if the file doesn't exist
10 try:
11     with open(csv_filename, 'x', newline='') as csvfile:
12         csv_writer = csv.writer(csvfile)
13         csv_writer.writerow(csv_header)
14 except FileExistsError:
15     pass # File already exists, no need to create header again
16
17 def find_rfid_data(rfid):
18     with open(csv_filename, 'r', newline='') as csvfile:
19         csv_reader = csv.reader(csvfile)
20         for row in csv_reader:
21             if rfid in column:
22                 return column
23     return None
24
25 print("Please insert an RFID card")

```

Fig. 6: Code snippet

```

        return column
    return None

print("Please insert an RFID card")

while True:
    card = reader.read()
    if card:
        rfid = card.value
        existing_data = find_rfid_data(rfid)

        if existing_data:
            name, date, time = existing_data[1:4]
            print("RFID {} already exists with name {} on {} at {}".format(rfid, name, date, time))
        else:
            name = input("Enter a name for the RFID {}: ".format(rfid))
            date = datetime.datetime.now().strftime('%Y-%m-%d')
            time = datetime.datetime.now().strftime('%H:%M:%S')

            with open(csv_filename, 'a', newline='') as csvfile:
                csv_writer = csv.writer(csvfile)
                csv_writer.writerow([rfid, name, date, time])

            print("RFID {} saved with name {} on {} at {}".format(rfid, name, date, time))

```

Fig. 7: Loop snippet

it in a computer or cloud servers. Despite the chip being inside the body, automatic detection is essential if animals or snakes escape

their enclosures, necessitating the design of a reader device capable of RFID detection from a distance. Refer to Fig. 1 for the chip's location

## BILL OF MATERIALS

| Components               | Quantity | Description              |
|--------------------------|----------|--------------------------|
| Implantable RFID capsule | 1        | 125Hz                    |
| RDM 6300 with ANT.1      | 1        | 125Hz serial RFID reader |
| Raspberry Pi Zero / 4    | 1        | SBC                      |
| Jumper wire              | 10       | Female-female            |
| 5V/ 2A DC adaptor        | 1        | For powering             |
| USB FTDI                 | 1        | For serial reading       |

inside the body seen using X-ray. You may commence the project with the collection of components (bill of materials) detailed in the table above.

## Circuit and working

The circuit diagram of the device is shown in Fig. 2. The circuit comprises the 125Hz serial RFID reader chip RDM 6300 (MOD1), USB FTDI (MOD2) for interfacing with Raspberry Pi Zero/4, and a few other components. The antenna comes free with the RFID reader with proper specifications.

When it comes to tracking, not all microchip implants are equal. In this case, we leverage the power of a 125kHz RFID implantable device for animal and human identification.

This tiny yet powerful embedded chip acts as a digital fingerprint with crucial details such as age, health, species, and origin.

Reading and identifying these chips is straightforward; you can use a Raspberry Pi (RPI) or a laptop with Python and a USB connec-